

Supporting Information 1: R package usage

2026-07-17

This vignette contains the code and figures for comparing the relative usage of the R packages `gllvm` (Niku et al. 2025), `hmsc` (Tikhonov et al. 2025), `boral` (Hui 2025) and `ecoCopula` (Popovic et al. 2019), which are all bespoke model-based ordination packages, with the `vegan` package (Oksanen et al. 2025), a go-to for classical ordination, based on monthly downloads of each package from the Comprehensive R Archive Network (CRAN). In order to not confound comparison with other types of models, we did not include other general-purpose packages also capable of fitting GLLVMs, such as `glmmTMB` (McGillycuddy et al. 2025) and `VGAM` (Yee 2025).

Data setup

```
# load required R packages
library(cranlogs) # get package download data from cran
library(dplyr) # data handling
library(lubridate) # aggregate data by date
library(ggplot2) # plotting
```

We begin by loading the data, from the beginning of 2020 to the end of 2025.

```
data_downloads <- cran_downloads/packages=c("vegan","gllvm","Hmsc","boral",
                                             "ecoCopula"),
                      from = "2020-01-01", to = "2025-12-31")
downloads_monthly <- data_downloads |>
  group_by(month=floor_date(date, "month"),package) |>
  summarise(count=sum(count))
```

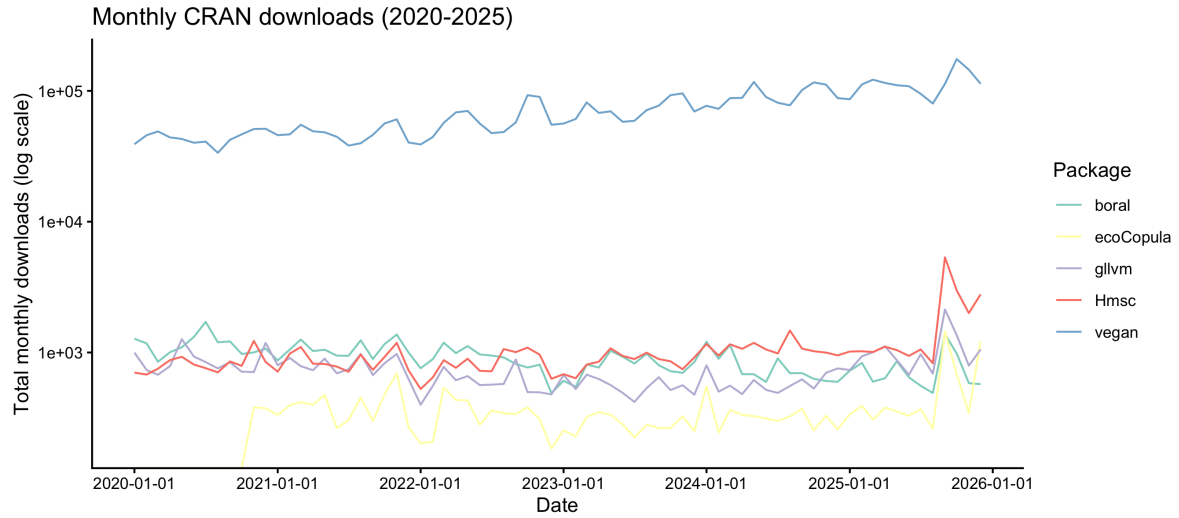
We also calculate the monthly ratio of downloads for each package:

```
downloads_monthly <- downloads_monthly |> mutate(ratio=count/sum(count))
```

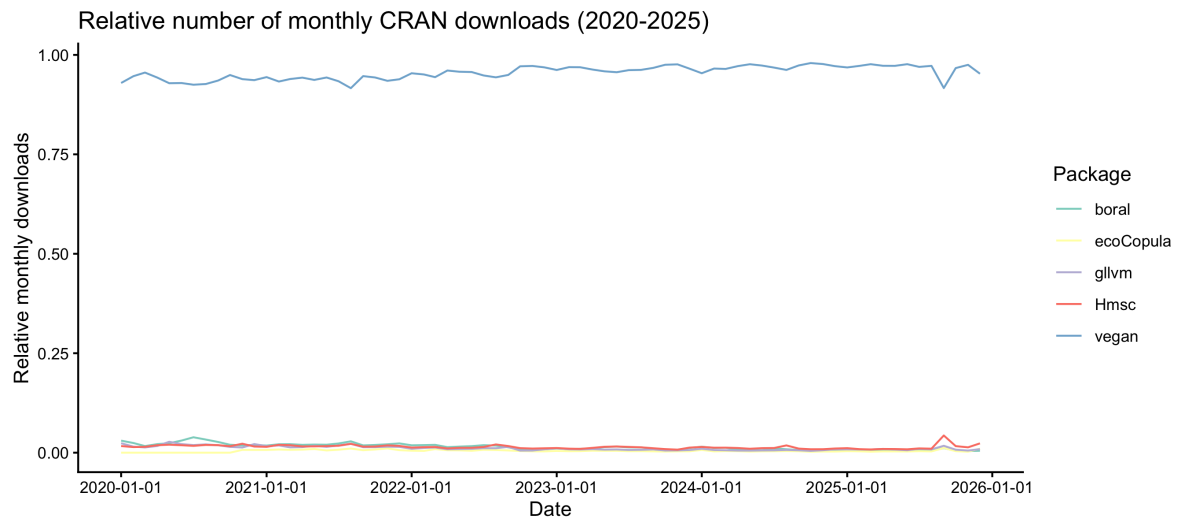
Plotting

We then plot the data in a graph:

```
ggplot(downloads_monthly) + # total downloads/month
  geom_line(aes(x=month, y=count, color=package),stat="identity") +
  scale_y_continuous(trans='log10') +
  theme_classic() +
  labs(color="Package", y="Total monthly downloads (log scale)", x="Date",
        title="Monthly CRAN downloads (2020-2025)") +
  scale_color_brewer(palette="Set3") +
  scale_x_date(date_breaks = "1 year")
ggplot(downloads_monthly) + # relative downloads/month
  geom_line(aes(x=month, y=ratio, color=package),stat="identity") +
  labs(color="Package", y="Relative monthly downloads", x="Date",
        title="Relative number of monthly CRAN downloads (2020-2025)") +
  scale_color_brewer(palette="Set3") +
  theme_classic() +
  scale_x_date(date_breaks = "1 year")
```



(a) Total number of monthly citations for each package (log scale)



(b) Percentage of total citations across all packages per month

Supplementary Figure 1. 1: yearly citations for the vegan R package (classical ordination tools) and R packages in which model-based ordination tools are central (boral, gllvm and hmsc).

References

- Hui, Francis K. C. 2025. *Boral: Bayesian Ordination and Regression AnaLysis*. <https://doi.org/10.32614/CRAN.package.boral>.
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- Niku, Jenni, Wesley Brooks, Riki Herliansyah, et al. 2025. *Gllvm: Generalized Linear Latent Variable Models*. <https://doi.org/10.32614/CRAN.package.gllvm>.
- Oksanen, Jari, Gavin L. Simpson, F. Guillaume Blanchet, et al. 2025. *Vegan: Community Ecology Package*. <https://doi.org/10.32614/CRAN.package.vegan>.
- Popovic, Gordana C., David I. Warton, Fiona J. Thomson, and Angela T. Moles. 2019. “Untangling Direct Species Associations from Indirect Mediator Species Effects with Graphical Models.” *Methods in Ecology and Evolution* 10 (9): 1571–83. <https://doi.org/10.1111/2041-210X.13247>.
- Tikhonov, Gleb, Otso Ovaskainen, Jari Oksanen, Melinda de Jonge, Oystein Opedal, and Tad Dallas. 2025. *Hmsc: Hierarchical Model of Species Communities*. <https://doi.org/10.32614/CRAN.package.Hmsc>.
- Yee, T. W. 2025. *VGAM: Vector Generalized Linear and Additive Models*. <https://doi.org/10.32614/CRAN.package.VGAM>.